



NATIONAL UNIVERSITY OF MODERN LANGUAGES

SOFTWARE QUALITY ENGINEERING LAB REPORT – 7

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COURSE: SOFTWARE QUALITY ENGINEERING

SUBMITTED TO: Mam Rubab

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Lab Report: Software Reviews and Fagan Inspection Process

Case Study: ATM System Requirement Specification

This report analyses the requirement specification of the ATM System and applies the Fagan Inspection Process to identify defects and improve software quality.

Task 1: Applying Fagan Inspection Process on ATM Requirement Specification

The Fagan Inspection Process is a formal review technique used to identify defects in software documents before implementation. The following phases were applied to inspect the ATM System requirement specification.

Phase	Description	Activities Performed
Planning	The moderator plans the inspection process.	Inspection team selected and SRS document distributed.
Overview Meeting	Requirements are explained to inspection team members.	Team reviewed ATM functional and non-functional requirements.
Preparation	Inspectors individually study the document.	Requirements checked for ambiguity, completeness, and consistency.
Inspection Meeting	Defects are discussed and recorded.	Missing details and unclear requirements were identified.
Rework	Author corrects the identified defects.	Requirements updated according to inspection feedback.
Follow-Up	Moderator verifies that defects are corrected.	Final review performed before approval.

Task 2: Defect Report According to Fagan Inspection Criteria

Defect ID	Requirement No.	Defect Description	Defect Type	Severity
D-01	1	Withdrawal limit does not specify daily transaction limit.	Incomplete Requirement	Medium
D-02	2	PIN verification process does not mention maximum retry attempts.	Missing Information	High
D-03	3	Currency format is not clearly defined for balance display.	Ambiguous Requirement	Low
D-04	4	Requirement does not specify behavior if printer is unavailable.	Incomplete Requirement	Medium
D-05	6	System response time requirement lacks testing conditions.	Unclear Requirement	Medium
D-06	7	Concurrent users requirement does not specify peak load duration.	Incomplete Requirement	Low

Task 3: Review Simulation for a Software Project

Selected Project:

Student Management System

Selected Artifact:

Software Requirement Specification (SRS)

Selected Review Type:

Formal Technical Review

Role	Responsibility	Phase Involved
Moderator	Organizes review meetings and manages process.	Planning and Follow-Up
Author	Explains document and fixes defects.	Overview and Rework
Reviewer	Examines document to identify defects.	Preparation and Inspection
Recorder	Documents all identified defects during meeting.	Inspection Meeting

Task 4: Sample Defect Log

Defect No.	Description	Severity	Status	Assigned To
1	Missing validation rule for student registration.	High	Open	Author
2	Incorrect field name in requirement document.	Low	Resolved	Author
3	Ambiguous attendance calculation requirement.	Medium	In Progress	Reviewer
4	Missing error handling requirement.	Medium	Open	Author

Task 5: Review Metrics Analysis

The following review metrics were calculated based on the inspection activity.

Metric	Value	Explanation
Number of Defects Found	6	Total defects identified during inspection.
Defects per Page	1.5	6 defects found in 4 pages of SRS document.
Review Effectiveness	75%	Most major defects were identified during review process.

Conclusion

The Fagan Inspection Process helped identify missing, unclear, and incomplete requirements in the ATM System requirement specification. Review activities and inspection metrics improved document quality and reduced the possibility of defects during software development.